

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

Name	Abinesh M
Register Number	192465050
Date	05-08-2026
Time	1 Hour

COURSE NAME: Cybersecurity Incident Handling And Response
COURSE CODE: CSA6001

COURSE OUTCOME COVERED

CO2: *Analyze incident response preparation strategies through examination of team roles, policies, SOPs, risk management practices, along with compliance frameworks to ensure organizational readiness. (BL3, BL6)*

Activity Tool : Incident Response Team Planning and SOP Development (High)
Assessment Tool Weightage: 40%

Code of Conduct:

*I Abinesh M 192465050 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

***Abinesh M
192465050***

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Incident Response Team Planning	20%	Clearly identifies all key Incident Response Team roles, responsibilities, reporting structure, and interdepartmental coordination.	Identifies most roles and responsibilities with minor omissions.	Identifies only some roles with limited understanding of responsibilities.	Unable to identify appropriate team roles or responsibilities.
SOP Development and Incident Handling Process	25%	Develops a comprehensive, logical SOP covering detection, reporting, containment, eradication, recovery, documentation, and communication.	SOP includes most required phases with minor omissions or sequencing errors.	SOP covers only basic steps with several missing components.	SOP is incomplete, incorrect, or lacks logical flow.
Security Policies, Awareness, and Governance	20%	Presents well-structured security policies, awareness program, and governance practices with clear justification.	Includes relevant policies and awareness activities with minor gaps.	Provides limited policies or awareness measures with insufficient explanation.	Fails to identify appropriate policies or governance practices.
Risk Assessment, BCP, and DRP	20%	Accurately identifies critical assets, performs risk assessment, and clearly explains BCP and DRP strategies with appropriate recommendations.	Performs basic risk assessment and explains BCP/DRP with minor omissions.	Partial risk assessment with limited understanding of BCP and DRP concepts.	Unable to perform risk assessment or explain BCP/DRP.
Documentation, Communication, and Presentation	15%	Documentation is well-organized, professionally presented, uses diagrams/tables effectively, and communicates ideas clearly.	Documentation is organized with minor formatting or clarity issues.	Documentation is partially organized with several presentation issues.	Documentation is poorly organized, incomplete, or difficult to understand.

Cybersecurity Incident Handling and Response

CO2 AT1 REPORT

Student	Abinеш M
Register Number	192465050
Course Code	CSA6001
Course Name	Cybersecurity Incident Handling and Response
Assessment	CO2 AT1

1. Resource Planning & Readiness

Overview

Resource planning ensures that an organization has the right people, technology and budget before a cyber incident occurs. Readiness assessments reveal weaknesses in processes, monitoring and communication so that improvements can be prioritized. A mature organization responds faster, reduces downtime and protects business operations. Resource planning ensures that an organization has the right people, technology and budget before a cyber incident occurs. Readiness assessments reveal weaknesses in processes, monitoring and communication so that improvements can be prioritized. A mature organization responds faster, reduces downtime and protects business operations.

Key Points

- Review existing security policies and incident response plan.
- Perform vulnerability assessment and penetration testing.
- Identify security gaps using audits and tabletop exercises.
- Evaluate employee awareness through phishing simulations.
- Assess compliance with NIST SP 800-61 and ISO/IEC 27035.

Aspect	Details
Benefit	Improves detection, coordination and faster recovery.
Challenges	Budget, skilled workforce and integration complexity.
Recommendation	Regular drills, monitoring and continuous improvement.

2. Human, Technical & Financial Resources

Overview

Successful incident response depends on balanced investment in people, technology and operational funding. Skilled analysts require reliable tools, while management must ensure continuous training and maintenance. Successful incident response depends on balanced investment in people, technology and operational funding. Skilled analysts require reliable tools, while management must ensure continuous training and maintenance.

Key Points

- SOC Analysts, Incident Responders, Digital Forensics Experts
- SIEM, EDR, IDS/IPS, Firewalls, Threat Intelligence
- Secure backup servers and disaster recovery infrastructure
- Training budget, software licensing and cyber insurance

Aspect	Details
Benefit	Improves detection, coordination and faster recovery.
Challenges	Budget, skilled workforce and integration complexity.
Recommendation	Regular drills, monitoring and continuous improvement.

3. SIEM & Ticketing Systems

Overview

SIEM platforms improve visibility by collecting logs from servers, cloud platforms and endpoints. Ticketing systems transform alerts into structured investigations, making collaboration and accountability much easier. SIEM platforms improve visibility by collecting logs from servers, cloud platforms and endpoints. Ticketing systems transform alerts into structured investigations, making collaboration and accountability much easier.

Key Points

- Centralized log collection
- Real-time event correlation
- Alert prioritization
- Incident assignment and SLA tracking
- Evidence documentation and reporting

Aspect	Details
Benefit	Improves detection, coordination and faster recovery.
Challenges	Budget, skilled workforce and integration complexity.
Recommendation	Regular drills, monitoring and continuous improvement.

4. Healthcare Ransomware Case Study

Overview

A ransomware attack on a hospital can delay patient care and disrupt critical operations. Asset prioritization ensures that life-saving systems are restored first while forensic investigations continue. A ransomware attack on a hospital can delay patient care and disrupt critical operations. Asset prioritization ensures that life-saving systems are restored first while forensic investigations continue.

Key Points

- Identify EHR databases and patient records
- Prioritize diagnostic systems and medical devices
- Isolate infected machines
- Validate clean offline backups
- Restore services in order of business impact

Aspect	Details
Benefit	Improves detection, coordination and faster recovery.
Challenges	Budget, skilled workforce and integration complexity.
Recommendation	Regular drills, monitoring and continuous improvement.

5. Risk Assessment, BCP & DRP

Overview

Business Continuity Planning keeps essential healthcare services operational during an incident, whereas Disaster Recovery Planning focuses on restoring IT systems and data after containment. Together they reduce downtime and improve organizational resilience. Business Continuity Planning keeps essential healthcare services operational during an incident, whereas Disaster Recovery Planning focuses on restoring IT systems and data after containment. Together they reduce downtime and improve organizational resilience.

Key Points

- Classify risks as High, Medium and Low
- Apply MFA, network segmentation and patch management
- Maintain offline immutable backups
- Use Business Continuity Planning for essential services
- Use Disaster Recovery Planning to restore infrastructure

Aspect	Details
Benefit	Improves detection, coordination and faster recovery.
Challenges	Budget, skilled workforce and integration complexity.
Recommendation	Regular drills, monitoring and continuous improvement.